

INSOPIN 7.5MG TABLET

DESCRIPTION: A yellow oval shaped tablet, scored on one side with marking '861' and marking 'DUO' on the other side.

COMPOSITION: Each tablet contains Zopiclone BP 7.5 mg

PHARMACODYNAMICS: Zopiclone, a cyclopyrrolone derivative, is a short-acting hypnotic agent. Zopiclone belongs to a novel chemical class which is structurally unrelated to existing hypnotics. The pharmacological profile of zopiclone is similar to that of the benzodiazepines.

PHARMACOLOGY: In sleep laboratory studies of 1 to 21-day duration in man, zopiclone reduced sleep latency, increased the duration of sleep and decreased the number of nocturnal awakenings. Zopiclone delayed the onset of REM sleep but did not reduce consistently the total duration of REM periods. The duration of stage 1 sleep was shortened, and the time spent in stage 2 sleep increased. In most studies, stage 3 and 4 sleep tended to be increased, but no change and actual decreases have also been observed. The effect of zopiclone on stage 3 and 4 sleep differs from that of the benzodiazepines which suppress slow wave sleep. The clinical significance of this finding is not known.

PHARMACOKINETICS:

Absorption, Distribution and Bioavailability: Zopiclone is rapidly absorbed and distributed after oral administration, the time of maximum observed plasma concentration being about 1.75 hours. A study of 16 healthy volunteers receiving a single dose of 7.5 mg of zopiclone intravenously demonstrated the apparent volume of distribution of zopiclone to be 104 ± 15.5 L.

Autoradiographic studies in the rat showed rapid distribution into the blood and peak tissue levels at 0.5 hours in the liver, small intestines, stomach, kidneys and the adrenals. After twenty four hours the total residual radioactivity in the body of the rat was 8%.

The bioavailability of the 7.5 mg tablets in man is $76.3 \pm 9.6\%$, a hepatic first pass effect has been demonstrated.

In fresh human plasma, zopiclone is approximately 45% protein bound in the 25-100 ng/mL concentration range.

Metabolism: Zopiclone is extensively and rapidly metabolised by the liver. A large number of metabolites have been isolated and characterised, with the two major ones being the N-oxide, produced by oxidation of the piperazine nitrogen and the N-desmethyl produced by oxidative demethylation of the N-methyl piperazine.

Only the N-oxide analogue has weak pharmacological activity.

Elimination and Excretion: Zopiclone is rapidly eliminated, mainly by means of hepatic metabolism. The elimination half-life after a single oral dose is 5.26 ± 0.76 hours. The elimination half-life for the N-oxide metabolite is 7.28 ± 0.49 hours and that for the N-desmethyl metabolite is 7.28 ± 0.49 hours.

Renal clearance is 13.9 ± 7.0 mL/min which further shows that the major elimination pathway is by hepatic metabolism.

The amount of renal excretion is also low, unchanged zopiclone 3.6%, the N-oxide metabolites 11.4% and the N-desmethyl metabolite 13.4%.

Elderly: In elderly patients, the absolute bioavailability is increased (94% vs 77% in young subjects), and the elimination half-life prolonged (approximately 7 hours).

Hepatic insufficiency: In patients with hepatic insufficiency, elimination half-life is prolonged (11.9) and time to peak plasma levels delayed (3.5 hours).

Renal insufficiency: In patients with mild to moderate renal insufficiency, the pharmacokinetics of zopiclone are not altered. Haemodialysis does not appear to increase the plasma clearance of the drug.

INDICATIONS: Short term treatment of insomnia (2 to 4 weeks).

RECOMMENDED DOSAGE:

Adults: 7.5 mg by oral administration shortly before retiring for a maximum of 2-4 weeks. This dose should not be exceeded. Depending on clinical response, the dose may be lowered to 3.75 mg.

Zopiclone is not recommended for long term use (i.e. periods of more than 4 weeks). If used for long periods, treatment should be withdrawn gradually (see Warnings).

Elderly patients: In the elderly and/or debilitated patient an initial dose of 3.75 mg is recommended. The dose may be increased to a maximum of 7.5 mg if the starting dose does not offer adequate therapeutic effect, but in clinical trials, 25% of elderly patients treated with zopiclone experienced CNS side effects at the higher dose. Zopiclone should be used with caution in these patients.

Paediatric: Zopiclone is contraindicated in children. Dosage has not been established.

Hepatic Insufficiency: The recommended dose is 3.75 mg depending on acceptability and efficacy. Up to 7.5 mg may be used with caution in appropriate cases.

Alternative therapy: For long term treatment of insomnia, alternative non-pharmacological methods should be considered. Effective practical management of insomnia must respond to the presenting characteristics of the complaint. Giving accurate information is a form of treatment; there is benefit in discussing some simple facts with the patient and relating them to the problem, thereby assisting the patient to place the sleep problem in its context. Sleep hygiene such as reduction of caffeine intake, should be exercised. Programmes designed to establish an optimal sleeping pattern for the patient may also be useful as are relaxation techniques designed to assist the patient deal with tension and intrusive thoughts in bed.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS: Patients with known hypersensitivity to zopiclone or any of the excipients, myasthenia gravis, severe impairments of respiratory function, acute cerebrovascular accident, sleep apnoea syndrome. Zopiclone is contraindicated in children.

WARNING & PRECAUTIONS: This product may be habit forming on prolonged use. Safety in children has not been established.

As with all patients taking CNS depressant medications, patients receiving zopiclone should be warned not to operate dangerous machinery or motor vehicles until it is known that they do not become drowsy after zopiclone therapy. Abilities may be impaired on the day following use. Patients should be advised that their tolerance for alcohol and other CNS depressants will be diminished and that these medications should be either eliminated or given in reduced dosage in the presence of zopiclone.

Dependence and Withdrawals: Experiments in monkeys have shown intermediate withdrawal signs at doses of 16 mg/kg PO bid to 32 mg/kg PO bid.

A risk of withdrawal phenomenon after abrupt discontinuation of zopiclone following prolonged use in patients cannot be excluded. It is therefore recommended that after prolonged use the dose should be decreased gradually and the patient advised about such a possibility.

Zopiclone should be prescribed for short periods only (2-4 weeks). Continuous long term use is not recommended. There have been reports of withdrawal symptoms and difficulty in stopping treatment after long term therapy. Prolonged use of hypnotics is not recommended especially in the elderly.

Anaphylaxis (Severe allergic reaction) and angioedema (severe facial swelling), which can occur as early as the first time the product is taken.
Complex Sleep – Related behaviors which may include sleep driving, making phone calls, preparing and eating food while asleep.

Hepatic and renal insufficiency: In patients with severe hepatic insufficiency (serum albumin less than 30 g/L or presence of gross oedema), the elimination of zopiclone may be significantly reduced. Treatment should be initiated on a dose of 3.75 mg and if necessary, should be carried out at 7.5 mg. Zopiclone is removed by dialysis.

Respiration: Caution should be exercised in treating patients with chronic respiratory insufficiency. Treatment should be initiated on a dose of 3.75 mg and if necessary, should be carried out at 7.5 mg.

Hormonal Systems: Treatment of rats with zopiclone increases hepatic thyroid hormone metabolism of T4, resulting in increases in TSH and T3 levels, and decreases in T4 levels. It is suggested that zopiclone should not be administered to individual with impaired thyroid hormone homeostatic mechanisms or with conditions linked to hormonal imbalances.

Depression, Psychosis and Schizophrenia: Caution should be exercised if zopiclone is prescribed to depressed patients, including those with latent depression, particularly when suicidal tendencies may be present and protective measures may be required.

Geriatric or Debilitated Patients: Such patients may be particularly susceptible to the sedative effects of zopiclone and associated giddiness, ataxia and confusion, which may increase the possibility of a fall.

Epilepsy: Patients with a history of seizures should not be abruptly withdrawn from any CNS depressant drug, including zopiclone.

Abuse: Caution must be exercised in administering zopiclone to individuals known to be addiction prone or those whose history suggests they may increase the dosage on their own initiative.

INTERACTION WITH OTHER MEDICATIONS: Additive CNS depressant effects should be expected if zopiclone is administered concomitantly with other medications which themselves produce CNS depression, for example, barbiturates, benzodiazepines, alcohol, sedatives, tricyclic antidepressants, non-selective MAO inhibitors, phenothiazines and other antipsychotics, skeletal muscle relaxants, antihistamines, narcotic analgesics and anaesthetics.

USE IN PREGNANCY & LACTATION:

Use in pregnancy: Studies in animals have not shown evidence of an increased occurrence of foetal damage. However, zopiclone has been shown to cross the placenta, and increase postnatal mortality in rats given 10 mg/kg/d and above. Although the significance of this for humans is not known, it is likely that zopiclone may be harmful to the neonate. Nevertheless, the established medical principle of prescribing medications during pregnancy only when absolutely indicated should be observed.

Use in Lactation: Zopiclone and/or its metabolites are excreted in breast milk so therefore use in nursing mothers is not recommended.

SIDE EFFECTS: The side effects most commonly seen in clinical trials is taste alteration (bitter taste).

More common reactions: *Gastrointestinal:* bitter taste, dry mouth. *Nervous system:* drowsiness, headaches, fatigue

Less common reactions: *Gastrointestinal:* heartburn, constipation, diarrhoea, nausea, coated tongue, bad breath, anorexia or increased appetite, vomiting, epigastric pains. *Nervous system:* agitation, anxiety, loss of memory including retrograde amnesia, confusion, dizziness, weakness, somnolence, asthenia, feeling of drunkenness, euphoria, depression, coordination abnormality, hypotonia, speech disorders, hallucinations (auditory and visual) behavioral disorders, aggression, tremor, rebound insomnia. *Cardiovascular:* palpitations in elderly patients. *Skin:* urticaria, tingling. *Miscellaneous:* blurred vision, micturition. *Reproductive:* impotence, ejaculation failure.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: Overdose of zopiclone can be manifested by drowsiness, lethargy and ataxia. To date, no serious effects have been seen in patients with overdose.

Treatment: Following overdosage with zopiclone, gastric lavage should be performed. Treatment should be supportive and in response to clinical signs and symptoms. Respiration, pulse and blood pressure should be monitored. Flumazenil may be useful as an antidote. As in the management of overdosage with any medication, it should be borne in mind that multiple agents may have been taken.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE: 100's tablets in blister pack.

SHELF LIFE: Please refer to outer package

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599 Jalan Seruling 59 Kawasan 3, Taman Klang Jaya,
41200 Klang, Selangor, MALAYSIA.